

Rotax 912ULS

Austrian company Rotax's 912 broke the mold for light aircraft engines when it was introduced in 80 hp form in 1989. While it resembled the air-cooled Volkswagen Beetle engine often converted for aircraft use, the higher-revving 912 had individual, water-cooled cylinder heads, allowing it to produce much more power for little penalty in weight.

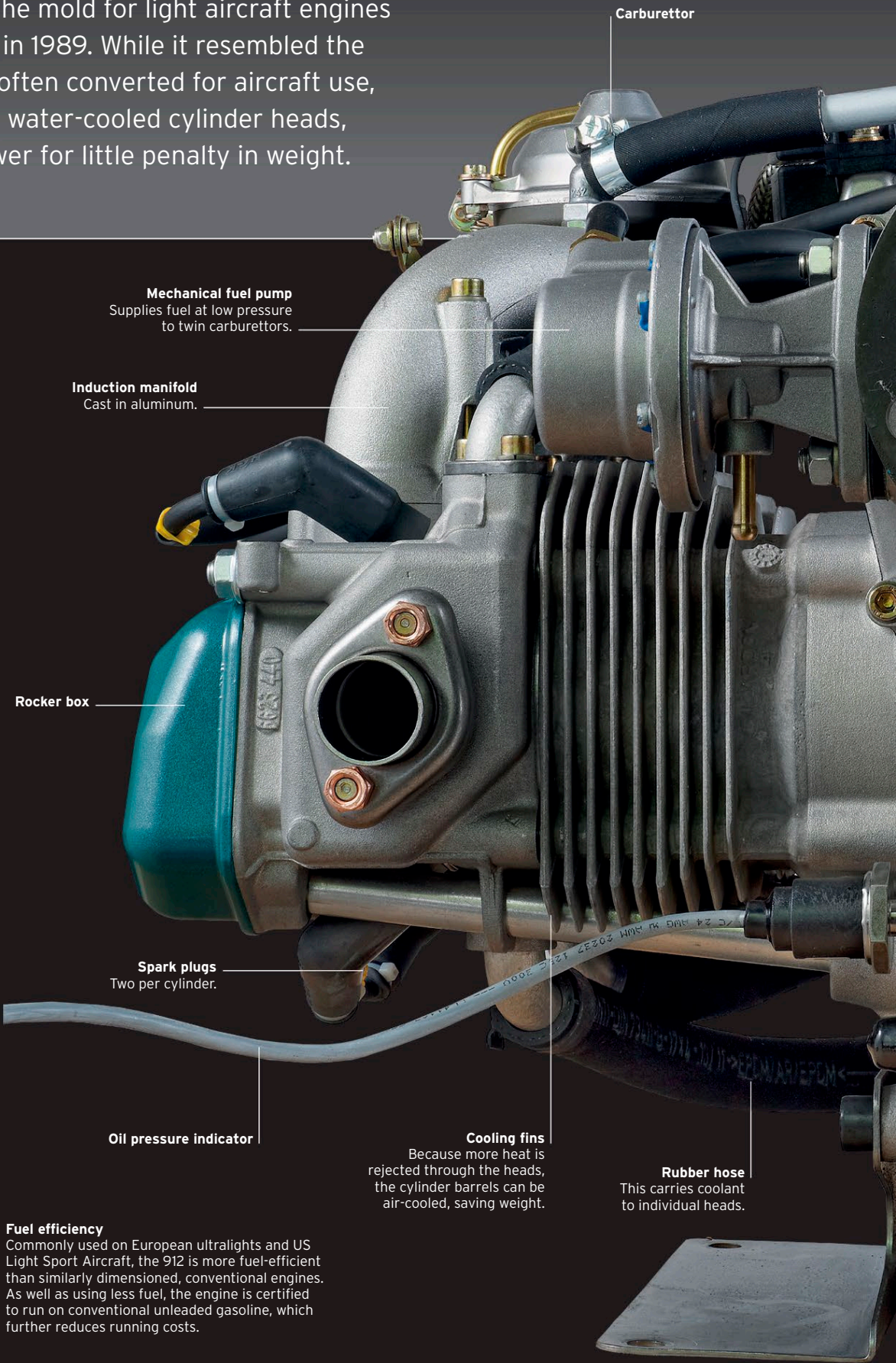
GEAR DRIVE HOLDS THE KEY

Because they did not have liquid-filled cooling systems that added weight and were often leak-prone, air-cooled engines held sway in the light aircraft world until the late 1980s. From the 1930s, US manufacturers had dominated the scene with simple, slow-revving, direct-drive engines that produced adequate power by virtue of relatively high capacity—the popular 100 hp Continental O-200 runs at 2,750 rpm and has a displacement of 201.3 cu in. Through developing a reliable propeller gear drive, Rotax was able to create the same power from a much smaller engine running at high revs, producing a much more efficient overall package.

ENGINE SPECIFICATIONS	
Dates produced	1989 to present
Configuration	Liquid/air-cooled opposed 4-cylinder 4-stroke
Fuel	Unleaded mogas or avgas
Power output	100 hp @ 5,800 rpm
Weight	132 lb (60kg) dry
Displacement	82.6 cu in (1.35 liters)
Bore and stroke	3.31 in x 2.4 in (84 mm x 61 mm)
Compression ratio	10:5:1



▷ See Piston Engines pp.302-303



Fuel efficiency
Commonly used on European ultralights and US Light Sport Aircraft, the 912 is more fuel-efficient than similarly dimensioned, conventional engines. As well as using less fuel, the engine is certified to run on conventional unleaded gasoline, which further reduces running costs.